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Inventor(s)

Chigullapalli; Sruti et al.

COMPUTING SYSTEM WITH EXTERNAL DISCONNECTS

Abstract

A computing device includes a plurality of rear inlet disconnects extend from the rear side of the housing. The rear inlet disconnects are operable to be fluidly coupled with a rack manifold and receive fluid from the rack manifold. The rear inlet disconnects include a rear-to-front disconnect and a rear-to-cooling disconnect. A front inlet manifold is coupled with the housing and includes at least one front inlet disconnect operable to be fluidly coupled with one or more inlet cooling disconnects extending from the front side of the housing. The inlet cooling disconnects are operable to be fluidly connected with the liquid cooling system. An inlet transmission tube is coupled with an outside surface of the housing. The inlet transmission tube in fluid communication with the rear-to-front disconnect and operable to direct the fluid from the rear-to-front disconnect to the front inlet manifold.

Inventors: Chigullapalli; Sruti (Seattle, WA), Anderson; Paul (Leander, TX), Russo; Frank (Weehawken, NJ)

Applicant: ZT GROUP INT'L, INC. DBA ZT SYSTEMS (SECAUCUS, NJ)

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Background/Summary

FIELD

[0001] The present disclosure relates generally to a computing system with a liquid cooling system that includes external disconnects (e.g., no internal disconnects).

BACKGROUND

[0002] Computing systems such as modular servers and/or information handling systems are continually improving and becoming more powerful. The chips, for example artificial intelligence chips, are becoming more powerful, which leads to increased thermal design power. To manage the heat generated by the chips, liquid cooling is becoming a main strain on the hyperscale server industry. For example, liquid heat capture rate of greater than 90% is a goal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Implementations of the present technology will now be described, by way of example only, with reference to the attached figures, wherein:

[0004] FIG. 1 illustrates a computing system;

[0005] FIG. 2A illustrates a rear view of a computing device;

[0006] FIG. 2B illustrates a front view of the computing device;

[0007] FIG. 2C illustrates a front view of the computing device, showing the computing components of the computing device;

[0008] FIG. 3 illustrates a front view of the computing device, with inlet conduits coupled with front inlet disconnects and inlet cooling disconnects;

[0009] FIG. 4 illustrates a drip bib coupled with the front outlet manifold and/or the plurality of rear inlet disconnects;

[0010] FIG. 5A illustrates the drip bib;

[0011] FIG. 5B illustrates an exploded view of the drip bib; and

[0012] FIG. 6 illustrates the drip bib coupled with an internal manifold.

DETAILED DESCRIPTION

[0013] It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale and the proportions of certain parts may be exaggerated to better illustrate details and features of the present disclosure.

[0014] Several definitions that apply throughout this disclosure will now be presented. The term “coupled” is defined as connected, whether directly or indirectly through intervening components, and is not necessarily limited to physical connections. The term “substantially” is defined to be essentially conforming to the particular dimension, shape or other word that substantially modifies, such that the component need not be exact. For example, substantially cylindrical means that the object resembles a cylinder, but can have one or more deviations from a true cylinder. The term “about” means reasonably close to the particular value. For example, about does not require the exact measurement specified and can be reasonably close. As used herein, the word “about” can include the exact number. The term “near” as used herein is within a short distance from the particular mentioned object. The term “near” can include abutting as well as relatively small

distance beyond abutting. The terms “comprising,” “including” and “having” are used interchangeably in this disclosure. The terms “comprising,” “including” and “having” mean to include, but not necessarily be limited to the things so described.

[0015] Liquid cooled computing devices are becoming the norm, for example for artificial intelligence and/or high performance computing space. Large computing components such as processing units (e.g., central processing units (CPUs), tensor processing units (TPUs), graphics processing units (GPUs), and/or computational architectures of any other suitable type) being liquid cooled can provide a 70% heat capture rate. To achieve greater than 90% heat capture rate, multiple computing components need to be liquid cooled. For conventional computing systems, this can come at the cost of serviceability (e.g., number of disconnects and their locations) and/or a greater number of leak points.

[0016] Processing units are positioned adjacent or closer to the rear side of the computing device while peripheral component interconnect express (PCIe) cards are positioned adjacent or closer to the front side of the computing device. Accordingly, higher heat capture rate requires cooling of the processing units as well as the PCIe cards.

[0017] The presently disclosed computing device provides two sets of disconnects (e.g., blind mate quick disconnects) in the rear side of the computing device out of which one set of the disconnects supplies fluid to the front PCIe cards. All liquid connections for the computing device occur outside of the housing of the computing device thereby eliminating all internal disconnects and possible leak points inside the computing device.

[0018] In at least one example, liquid drip bibs can be provided to catch any stray drops that occur during disconnect. For example, drip bibs can be coupled with manifolds of the computing device. Accordingly, the drip bibs can be operable to catch any fluid that leaks from the disconnects of the manifolds. In some examples, an absorbent sponge can be included in a tray of the drip bib(s) to eliminate any chance of fluid spillage from the drip bib(s).

[0019] The disclosure now turns to FIG. 1, which illustrates an example of a computing system **10** to be used, for example, with a liquid cooled modular server and/or information handling system. The computing system **10** includes a rack **5** with a rack manifold **6**. The rack **5** is operable to receive one or more computing devices **100**. The computing devices **100** can be fluidly coupled with the rack manifold **6** such that fluid is operable to flow between the rack manifold **6** and the computing device **100**, for example through a liquid cooling system for the computing device **100**.

[0020] FIG. 2A illustrates a rear view of the computing device **100**, and FIGS. 2B and 2C illustrates a front view of the computing device **100**. The computing device **100** can include a housing **12** that is operable to receive a plurality of computing components including a liquid cooling system operable to lower the temperature of the computing components. The housing **12** can include a front side **14**, a rear side **16** opposite the front side **14**, a left side **18** and a right side **20**. The left side **18** and the right side **20** can span between the front side **14** and the rear side **16**. When the computing device **100** is received in the rack **5**, the rear side **20** of the housing **12** can be positioned adjacent to the rack manifold **6**.

[0021] As shown in FIG. 2B, the computing device **100** can include a plurality of computing components **1, 2, 3, 4** received in the housing **12**. For example, the computing components can include one or more processing units **1, 2** (e.g., CPU, TPU, GPU, and/or any other suitable computational architecture) and one or more peripheral component interconnect express (PCIe) cards **3, 4**. At least one of the computing components, for example the processing unit **1** and/or the processing unit **2**, can be positioned in the housing **12** adjacent to the rear side **16** of the housing **12**. At least one of the computing components **3, 4**, for example the PCIe cards, rulers, hard drives, etc., can be positioned in the housing **12** adjacent to the front side **14** of the housing **12**. In some examples, the computing components **3, 4** (e.g., the PCIe cards, rulers, hard drives, etc.) positioned adjacent to the front side **14** of the housing **12** be received in one or more modules **108**, for example a left module **109** and a right module **110**. The left module **109** can be adjacent to the left

side **18** of the housing **12**, and the right module **110** can be adjacent to the right side **20** of the housing **12**. Left module **109** and the right module **110** can include separate connections to the liquid cooling system.

[0022] Thermal management of the computing systems **10** and the computing device **100** can be critical to the performance and time between failures for the computing system **10**. As computing devices **100** in computing systems **10** such as modular servers and/or information handling systems can have higher temperature environments, air cooling can be insufficient to adequately cool down the computing systems **10**. However, liquid cooling systems can bring about a cost of serviceability and leak point. High heat capture (e.g., heat capture rate of greater than 90%) can include cooling of the computing components **1, 2** (e.g., processing units **1, 2**) which are positioned adjacent to the rear side **16** of the housing **12** as well as the computing components **3, 4** (e.g., PCIe cards, rulers, hard drives, etc.) positioned adjacent to the front side **14** of the housing **12**. Increasing the number of computing components **1, 2, 3, 4** that need liquid cooling can increase the leak points which can damage the computing device **100**. Accordingly, to prevent leaking in the computing device **100**, the computing device **100** as disclosed herein does not use any internal quick disconnects (e.g., disconnects for liquid cooling fluid disposed inside the housing **12**). All of the disconnects are disposed outside of the housing **12** so that if there are any leaks, the computing components **1, 2, 3, 4** are not damaged. Additionally, as the left module **3** and the right module **4** are installed adjacent to the front side **14** of the computing device **100**, liquid cooling for the left module **3** and the right module **4** are connected at the front interface of the computing device **100**, which leads to improved ease of serviceability.

[0023] As illustrated in FIG. 2A, the computing device **100** includes a plurality of rear inlet disconnects **102, 104** extending from the rear side **16** of the housing **12**. The rear inlet disconnects **102, 104** are operable to be fluidly coupled with the rack manifold **6** and receive the fluid from the rack manifold **6**. The rear inlet disconnects **102, 104** can include a rear-to-front disconnect **104** and a rear-to-cooling disconnect **102**. The rear-to-front disconnect **104** can be operable to direct the fluid from the rack manifold **6** towards the front side **14** of the housing **12**. The rear-to-cooling disconnect **102** can be operable to be fluidly coupled with the liquid cooling system to lower the temperature of the computing components **1, 2** (e.g., the processing units **1, 2**) positioned adjacent to the rear side **16** of the housing **12**.

[0024] The computing device **100** can also include a plurality of rear outlet disconnects **114, 116** extending from the rear side **16** of the housing **12**. The rear outlet disconnects **114, 116** can be operable to be fluidly coupled with the rack manifold **6** and provide the fluid from the liquid cooling system of the computing device **100** back to the rack manifold **6**. The rear outlet disconnects **114, 116** can include a front-to-rear disconnect **114** and a cooling-to-rear disconnect **116**. The front-to-rear disconnect **114** can be operable to receive fluid from the front side **14** of the housing **12** and provide the fluid to the rack manifold **6**. The cooling-to-rear disconnect **116** can be operable to receive the fluid from the liquid cooling system after the fluid lowers the temperature of the computing components **1, 2** (e.g., the processing units **1, 2**) positioned adjacent to the rear side **16** of the housing **12**.

[0025] For example, the fluid can flow from the rack manifold **6** into the liquid cooling system for the computing device **100** via the rear-to-cooling disconnect **102**. The fluid flows in the liquid cooling system to the computing components **1, 2** adjacent to the rear side **16** of the housing **12** and lowers the temperature of the computing components **1, 2**. The fluid then flows out of the liquid cooling system of the computing device **100** through the cooling-to-rear disconnect **116** and into the rack manifold **6**.

[0026] In at least one example, the rear inlet disconnects **102, 104** and/or the rear outlet disconnects **114, 116** can include quick disconnects. The rear inlet disconnects **102, 104** and/or the rear outlet disconnects **114, 116** can be operable to automatically close upon disconnection of the rear inlet disconnects **102, 104** and/or the rear outlet disconnects **114, 116** with the rack manifold **6**. For

example, the rear inlet disconnects **102**, **104** and/or the rear outlet disconnects **114**, **116** can include valves such that when the rear inlet disconnects **102**, **104** and/or the rear outlet disconnects **114**, **116** are disconnected from the rack manifold **6**, the corresponding valve(s) and automatically close to prevent fluid from leaking in the computing system **10**. Accordingly, the computing device **100** can be disconnected from the rack manifold **6** such that the computing device **100** can be serviced with reduced amounts of fluid and/or no leaking fluid which can damage the computing system **10**. [0027] As the rear inlet disconnects **102**, **104** and the rear outlet disconnects **114**, **116** are positioned external to the housing **12**, no fluid connection is being made within the housing **12**. Accordingly, any possible leaks would not lead to fluid leaking into the housing **12** and impacting the computing components **1**, **2**, **3**, **4**. Additionally, serviceability of the computing device **100** is improved since the computing device **100** can be disconnected easily with external connections **102**, **104**, **114**, **116**. [0028] The computing device **100** includes a front inlet manifold **107** coupled with the housing **12**. The front inlet manifold **107** can be coupled with the front side **14** of the housing **12**, as illustrated in FIGS. 2A and 2B. The front inlet manifold **107** is operable to be fluidly connected with the rear-to-front disconnect **104** such that the fluid from the rack manifold **6** can flow from the rear-to-front disconnect **104** to the front inlet manifold **107**.

[0029] For example, as illustrated in FIGS. 2A-2C, an inlet transmission tube **106** can be in fluid communication with the rear-to-front disconnect **104** and the front inlet manifold **107** such that the inlet transmission tube **106** is operable to direct the fluid from the rear-to-front disconnect **104** to the front inlet manifold **107**. In at least one example, the inlet transmission tube **106** can be made at least partially from copper. In some examples, the inlet transmission tube **106** can be made of any suitable material to transport the fluid between the rear-to-front disconnect **104** and the front inlet manifold **107** without deviating from the scope of the disclosure. As illustrated in FIGS. 2A-2C, the inlet transmission tube **106** can be coupled with an outside surface **30** of the housing **12**. For example, the housing **12** can form an inlet groove **31** operable to receive the inlet transmission tube **106**. The inlet transmission tube **106** is positioned external to the housing **12** of the computing device **100** such that the fluid remains outside of the housing **12** in case of leaks and for easy serviceability. By being received in the inlet groove **31**, the footprint of the computing device **100** is not impacted or increased so that the computing device **100** can still be received in the rack **5** without needing any modification of rack **5** of the computing system **10**.

[0030] With the inlet transmission tube **106** passing from the rear-to-front disconnect **104** to the front inlet manifold **107**, the fluid can be directed to the front side **14** of the housing **12**, so that the fluid can flow to the computing components **3**, **4** positioned adjacent to the front side **14** of the housing **12**—without utilizing internal fluid connections.

[0031] In at least one example, as illustrated in FIGS. 2B-3, the front inlet manifold **107** can include at least one front inlet disconnect **200** operable to be fluidly coupled with one or more inlet cooling disconnects **210**, **212** extending from the front side **14** of the housing **12** via corresponding one or more inlet conduits **140**, **142**. The front inlet manifold **107** distributes the fluid received from the inlet transmission tube **106** between the at least one front inlet disconnect **200**. The inlet cooling disconnects **210**, **212** are operable to be fluidly connected with the liquid cooling system, for example to lower the temperature of the computing components **3**, **4** (e.g., PCIe cards) positioned adjacent to the front side **14** of the housing **12**. For example, the inlet cooling disconnects **210**, **210** are operable to be fluidly connected with the liquid cooling system to lower the temperature of the computing components **3** within the left module **109** and the computing components **4** within the right module **110** adjacent to the front side **14** of the housing **12**.

[0032] For example, as illustrated in FIGS. 2B-3, the front inlet manifold **107** can include a left front inlet disconnect **201** that is operable to be fluidly coupled with the left inlet conduit **140** that is operable to be fluidly coupled with the left inlet cooling disconnect **210**. The fluid can then enter the liquid cooling system for the left module **109** through the left inlet cooling disconnect **210**. In at least one example, the front inlet manifold **107** can also include a right front inlet disconnect **202**

that is operable to be fluidly coupled with the right inlet conduit **142** that is operable to be fluidly coupled with the right inlet cooling disconnect **212**. The fluid can then enter the liquid cooling system for the right module **110** through the right inlet cooling disconnect **212**. This can allow for liquid cooling for more computing components **1, 2, 3, 4**, leading to higher heat capture rate.

[0033] After lowering the temperatures of the computing components **3, 4**, one or more outlet cooling disconnects **220, 222** fluidly coupled with the liquid cooling system received in the housing **12** can be operable to receive the fluid from the liquid cooling system. The fluid can then flow to a front outlet manifold **111** towards the rear outlet disconnect **114** and the rack manifold **6**. The front outlet manifold **111** can include at least one front outlet disconnect **300** extending from the front side **14** of the housing **12**. The front outlet disconnects **300** can be fluidly coupled with the outlet cooling disconnects **114** via corresponding one or more outlet conduits **150, 152**. For example, as illustrated in FIGS. 2B and 2C, the left outlet cooling disconnect **220**—in communication with the liquid cooling system for the left module **108**—can be fluidly coupled with a left outlet conduit **150** which is fluidly coupled with the left front outlet disconnect **301** of the front outlet manifold **111**. The right outlet cooling disconnect **222**—in communication with the liquid cooling system for the right module **110**—can be fluidly coupled with a right outlet conduit **152** which is fluidly coupled with the right front outlet disconnect **302** of the front outlet manifold **111**.

[0034] In at least one example, the front inlet disconnects **200**, inlet cooling disconnects **210, 212**, outlet cooling disconnects **220, 222**, and/or the front outlet disconnects **300** can include quick disconnects. The front inlet disconnects **200**, inlet cooling disconnects **210, 212**, outlet cooling disconnects **220, 222**, and/or the front outlet disconnects **300** can be operable to automatically close upon disconnection of the front inlet disconnects **200**, inlet cooling disconnects **210, 212**, outlet cooling disconnects **220, 222**, and/or the front outlet disconnects **300** with the inlet conduits **140, 142** and/or the outlet conduits **150, 152**. For example, the front inlet disconnects **200**, inlet cooling disconnects **210, 212**, outlet cooling disconnects **220, 222**, and/or the front outlet disconnects **300** can include valves such that when the front inlet disconnects **200**, inlet cooling disconnects **210, 212**, outlet cooling disconnects **220, 222**, and/or the front outlet disconnects **300** are disconnected from the inlet conduits **140, 142** and/or the outlet conduits **150, 152**, the corresponding valve(s) and automatically close to prevent fluid from leaking in the computing system **10**. Accordingly, the computing device **100** can be serviced with reduced amounts of fluid and/or no leaking fluid which can damage the computing system **10**. Additionally, the computing components **3, 4** (e.g., in the left module **109** and the right module **110**) can be manually serviced from the front side **14** of the housing **12** without the need of turning around or removing the computing device **100** from the rack **5**.

[0035] As the front inlet disconnects **200**, inlet cooling disconnects **210, 212**, outlet cooling disconnects **220, 222**, and the front outlet disconnects **300** are positioned external to the housing **12**, no fluid connection is being made within the housing **12**. Accordingly, any possible leaks would not lead to fluid leaking into the housing **12** and impacting the computing components **1, 2, 3, 4**. Additionally, serviceability of the computing device **100** is improved since the computing device **100** can be disconnected easily with external connections **200, 210, 212, 220, 222, 300**.

[0036] The fluid can be directed from the front outlet manifold **111** to the front-to-rear disconnect **114** via an outlet transmission tube **112**. The outlet transmission tube **112** can be coupled with the outside surface of the housing **12**. The outlet transmission tube **112** can be in fluid communication with the front-to-rear disconnect **114** and operable to direct the fluid from the front outlet manifold **111** to the front-to-rear disconnect **114**. For example, the housing **12** can form an outlet groove **32** operable to receive the outlet transmission tube **112**. The outlet transmission tube **112** is positioned external to the housing **12** of the computing device **100** such that the fluid remains outside of the housing **12** in case of leaks and for easy serviceability. By being received in the outlet groove **32**, the footprint of the computing device **100** is not impacted or increased so that the computing device **100** can still be received in the rack **5** without needing any modification of rack **5** of the computing

system **10**. In at least one example, the outlet transmission tube **112** can be positioned along the right side **20** of the housing **12**, opposite the inlet transmission tube **106**.

[0037] FIGS. 4-5B illustrate a drip bib **400** that is operable to catch and receive the fluid that leaks from the disconnects (e.g., front inlet disconnects **200** and front outlet disconnects **300**). The drip bibs **400** can be operable to be coupled with the rear inlet disconnects **102**, **104**, front inlet manifold **107**, front outlet manifold **111**, and/or the rear outlet disconnects **114**, **116** to catch the leaked fluid.

[0038] As illustrated in FIGS. 4-5B, the drip bib(s) **400** can include a coupler **402** operable to couple with the rear inlet disconnects **102**, **104**, front inlet manifold **107**, front outlet manifold **111**, and/or the rear outlet disconnects **114**, **116**. For example, the coupler **402** can form an aperture **404** that is operable to receive each of the rear inlet disconnects **102**, **104**, front inlet manifold **107**, front outlet manifold **111**, and/or the rear outlet disconnects **114**, **116**. The aperture **404** can be, for example, a rectangular shape. In some examples, the coupler **402** can utilize friction fit, adhesion, etc. to couple with the rear inlet disconnects **102**, **104**, front inlet manifold **107**, front outlet manifold **111**, and/or the rear outlet disconnects **114**, **116**. The coupler **402** can be configured to be retrofit to the computing device **100** such that no modification is needed to the computing device **100** to utilize the drip bib(s) **400**.

[0039] For example, FIG. 6 illustrates the drip bib coupled with an internal manifold **602**. The internal manifold **602** may be operable to couple a plurality of internal conduits **600** that can permit fluid to flow therethrough. An internal disconnect **610** may extend from the internal manifold **602**. To prevent leaking of fluid onto any computing components, a drip bib **400** can be installed on the internal manifold **602** to receive the internal disconnect **610**. The coupler **402** of the drip bib **400** may include a hanger component **4020** that is operable to latch onto the internal manifold **602**.

[0040] The drip bib **400** can include a tray **406** operable to receive the fluid that leaks from the rear inlet disconnects **102**, **104**, front inlet manifold **107**, front outlet manifold **111**, and/or the rear outlet disconnects **114**, **116**. The tray **406** can form a recess **408** that is operable to retain the leaked fluid without overflow. In at least one example, for example as illustrated in FIGS. 5A and 5B, the drip bib **400** can include a sponge **500** that is operable to be received in the tray **406**. The sponge **500** can be operable to absorb the fluid received in the tray **406** to prevent overflow. For example, the sponge **500**, upon contact with the fluid (e.g., PG25), can convert to gel and can be safer to handle than using a rag to clean up the leaked fluid in the tray **406**.

[0041] In at least one example, the tray **406** can have a front surface **410** that includes an indentation **412**. The indentation **412** can be formed to correspond with the shape and/or size of the disconnects. Accordingly, the drip bib **400** can be easily installed and/or removed without needing any modification to the computing device **100**.

[0042] The embodiments shown and described above are only examples. Even though numerous characteristics and advantages of the present technology have been set forth in the foregoing description, together with details of the structure and function of the present disclosure, the disclosure is illustrative only, and changes may be made in the detail, especially in matters of shape, size and arrangement of the parts within the principles of the present disclosure to the full extent indicated by the broad general meaning of the terms used in the attached claims. It will therefore be appreciated that the embodiments described above may be modified within the scope of the appended claims.

Claims

1. A computing device comprising: a housing including a front side and a rear side opposite the front side; a plurality of rear inlet disconnects extending from the rear side of the housing, the plurality of rear inlet disconnects operable to be fluidly coupled with a rack manifold and receive fluid from the rack manifold, the plurality of rear inlet disconnects including a rear-to-front disconnect and a rear-to-cooling disconnect; a front inlet manifold coupled with the housing, the

front inlet manifold including at least one front inlet disconnect operable to be fluidly coupled with one or more inlet cooling disconnects extending from the front side of the housing via corresponding one or more inlet conduits, wherein the one or more inlet cooling disconnects are operable to be fluidly connected with the liquid cooling system; an inlet transmission tube coupled with an outside surface of the housing, the inlet transmission tube in fluid communication with the rear-to-front disconnect and operable to direct the fluid from the rear-to-front disconnect to the front inlet manifold.

2. The computing device of claim 1, wherein the one or more inlet cooling disconnects are operable to be fluidly connected with a liquid cooling system received in the housing.

3. The computing device of claim 2, wherein the one or more inlet cooling disconnects are operable to be fluidly coupled with the liquid cooling system to lower a temperature of one or more peripheral component interconnect express cards.

4. The computing device of claim 1, wherein the rear-to-cooling disconnect is operable to be fluidly coupled with the liquid cooling system to lower a temperature of one or more processing units.

5. The computing device of claim 1, wherein the housing forms an inlet groove operable to receive the inlet transmission tube.

6. The computing device of claim 1, further comprising one or more drip bibs coupled with the plurality of rear inlet disconnects and/or the front inlet manifold, each of the one or more drip bibs including a tray operable to receive the fluid that leaks from the plurality of rear inlet disconnect and/or the front inlet manifold.

7. The computing device of claim 6, wherein the one or more drip bibs includes a sponge operable to be received in the tray, the sponge operable to absorb the fluid received in the tray.

8. The computing device of claim 1, further comprising: one or more outlet cooling disconnects fluidly coupled with a liquid cooling system received in the housing, the one or more outlet cooling disconnects operable to receive the fluid from the liquid cooling system; and a front outlet manifold including at least one front outlet disconnect extending from the front side of the housing, wherein the at least one front outlet disconnect is fluidly coupled with the one or more outlet cooling disconnects via corresponding one or more outlet conduits.

9. The computing device of claim 8, further comprising: a plurality of rear outlet disconnects extending from the rear side of the housing, the plurality of rear outlet disconnects operable to be fluidly coupled with the rack manifold and provide the fluid to the rack manifold, wherein the plurality of rear outlet disconnects includes a front-to-rear disconnect and a cooling-to-rear disconnect; an outlet transmission tube coupled with the outside surface of the housing, the outlet transmission tube in fluid communication with the front-to-rear disconnect and operable to direct the fluid from the front outlet manifold to the front-to-rear disconnect.

10. The computing device of claim 9, wherein the cooling-to-rear disconnect is operable to receive the fluid from the liquid cooling system after the fluid lowers the temperature of one or more processing units positioned adjacent to the rear side of the housing.

11. The computing device of claim 9, further comprising one or more drip bibs coupled with the front outlet manifold and/or the plurality of rear outlet disconnects, each of the one or more drip bibs including a tray operable to receive the fluid that leaks from the at least one front inlet disconnect and/or the plurality of rear outlet disconnects.

12. The computing device of claim 11, wherein the one or more drip bibs includes a sponge operable to be received in the tray, the sponge operable to absorb the fluid received in the tray.

13. The computing device of claim 8, wherein the housing forms an outlet groove operable to receive the outlet transmission tube.

14. A computing system comprising: a rack manifold; a computing device fluidly coupled with the rack manifold, the computing device including: a housing including a front side and a rear side opposite the front side; a plurality of computing components; a liquid cooling system received in the housing, the liquid cooling system operable to lower a temperature of the plurality of

computing components; a plurality of rear inlet disconnects extending from the rear side of the housing, the plurality of rear inlet disconnect operable to be fluidly coupled with a rack manifold and receive fluid from the rack manifold, the plurality of rear inlet disconnects including a rear-to-front disconnect and a rear-to-cooling disconnect; a front inlet manifold coupled with the housing, the front inlet manifold including at least one front inlet disconnect operable to be fluidly coupled with one or more inlet cooling disconnects extending from the front side of the housing via corresponding one or more inlet conduits, wherein the one or more inlet cooling disconnects are operable to be fluidly connected with the liquid cooling system; an inlet transmission tube coupled with an outside surface of the housing, the inlet transmission tube in fluid communication with the rear-to-front disconnect and operable to direct the fluid from the rear-to-front disconnect to the front inlet manifold.

15. The computing system of claim 14, wherein the plurality of computing components include: one or more processing units, and one or more peripheral component interconnect express (PCIe) cards.

16. The computing system of claim 15, wherein the processing unit is disposed in the housing adjacent to the rear side of the housing, and wherein the one or more PCIe cards are disposed in the housing adjacent to the front side of the housing.

17. The computing system of claim 16, wherein the one or more PCIe cards positioned adjacent to the front side of the housing is in fluid communication, via the liquid cooling system, with the one or more inlet cooling disconnects such that fluid is operable to flow from the one or more inlet cooling disconnects into the liquid cooling system to the one or more PCIe cards to lower the temperature of the one or more PCIe cards.

18. The computing system of claim 16, wherein the one or more processing units positioned adjacent to the rear side of the housing is in fluid communication, via the liquid cooling system, with the rear-to-cooling disconnect such that fluid is operable to flow from the rear-to-cooling disconnect into the liquid cooling system to the one or more processing units to lower the temperature of the one or more processing units.

19. A method comprising: fluidly coupling a liquid cooling system of a computing device with a rack manifold via plurality of rear inlet disconnects extending from a rear side of a housing of the computing device, wherein the plurality of rear inlet disconnects includes a rear-to-front disconnect and a rear-to-cooling disconnect; directing flow of a fluid received by the rear-to-front disconnect from the rack manifold to a front inlet manifold coupled with a front side of the computing device via an inlet transmission tube coupled with an outside surface of the housing; and directing flow of the fluid from the front inlet manifold to the liquid cooling system for one or more front computing components positioned adjacent to a front side of the computing device through at least one front inlet disconnect fluidly coupled with one or more inlet cooling disconnects extending from the front side of the computing device via corresponding one or more inlet conduits.

20. The method of claim 19, further comprising: directing flow of the fluid received from the rack manifold by the rear-to-cooling disconnect to one or more rear computing components positioned adjacent to a rear side of the computing device.
